

MEMORANDUM

Class Test 2

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Electricity and Electronics EBN 122

20 September 2019

Test Information

Maximum marks:	16	Full marks:	15
Duration of test:	30 min + (10 min for OCR: 5 min before and 5 min at the end of the test)	Open/Closed book:	Closed
Total number of pages (this page included)		9	

Important

1. The examination regulations of the University of Pretoria apply.
2. Answer all questions on the OCR forms provided. The question number **01** in this question paper refers to the numeric field 01 of the OCR form, etc.
3. Read the instructions on the OCR form carefully.
4. Where applicable, answers must include units.
5. Final answers must be written as real numbers and not as fractions.
6. Use at least 3 significant numbers to write irrational numbers.

Lecturers: Prof J.P. de Villiers, Prof J.P. Jacobs, Dr X. Ye

INSTRUCTIONS

Do step 0 in the first 5 minutes of the session. (-5 to 0 min)

Do step 1 in the first 30 minutes of the test. (0-30 min)

Do step 2 within the following 5 minutes. (30-35 min)

STEP 0 (-5 min to 0 min): Complete the front page of the OCR form.

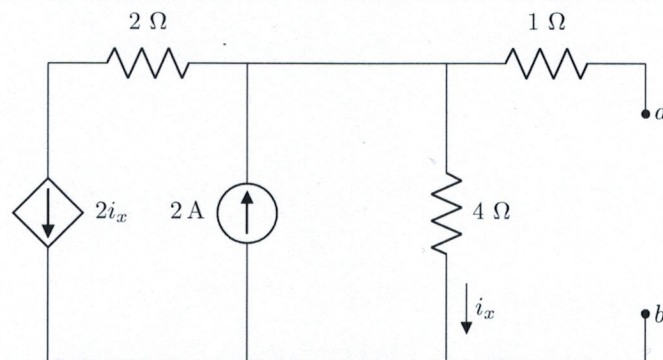
STEP 1 (0 min to 30 min): Do your own calculations on the question paper. Write down the answers for numeric questions on the PRACTICE ANSWER SHEET. Be sure to include the units where applicable.

STEP 2 (30 min to 35 min): (only if numeric sheets are completed) Carefully copy the answers from the PRACTICE ANSWER SHEET to the OCR form.

- The Assessment ID is: **2019EBN122C02**
- Use a mechanical pencil with 0.5 mm HB lead to complete the form.
- Each cell should only have one character, and characters may not touch or cross the cell.
- **Do not use commas!** Please make use of full stops as decimal points.

Section 1**Questions 1 to 2 [4]**

Find the Norton equivalent current I_N , and Norton equivalent resistance R_N , with reference to terminals $a - b$.

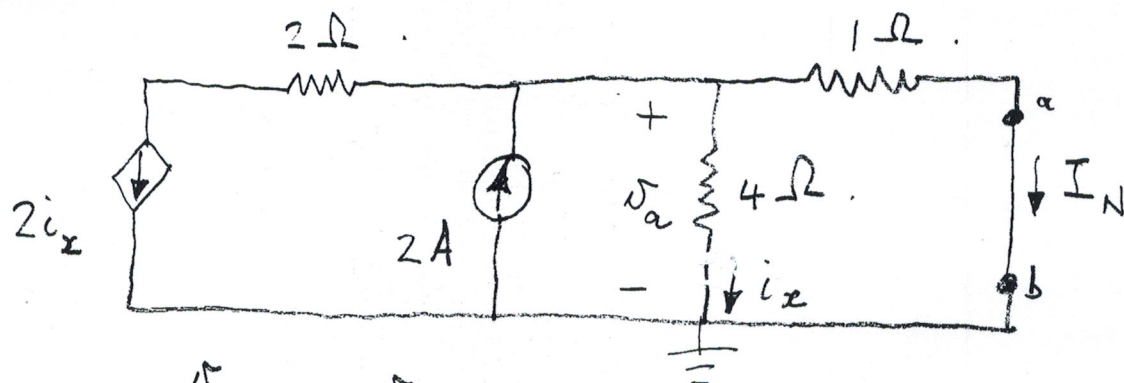


01 I_N [2]

02 R_N [2]

See next page

I_N :



KCL: $2i_x + \frac{v_a}{4} + \frac{v_a}{1} = 2$

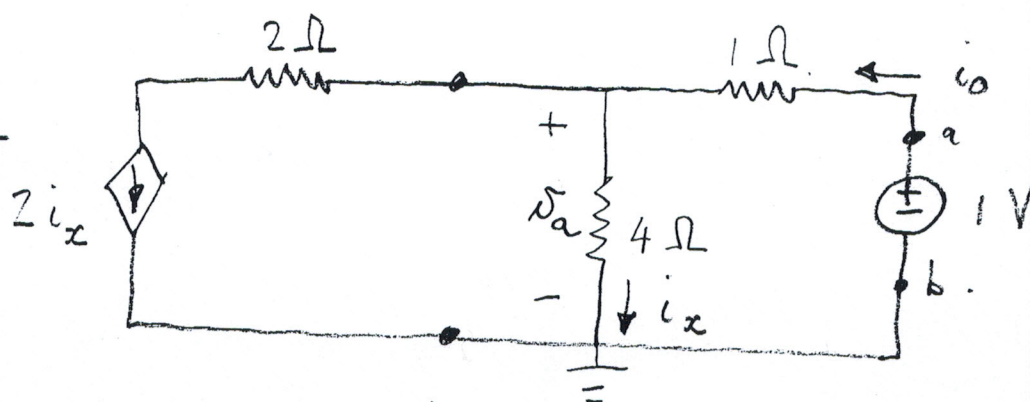
$$i_x = \frac{v_a}{4}$$

$$\therefore 2 \frac{v_a}{4} + \frac{v_a}{4} + v_a = 2$$

$$2v_a + v_a + 4v_a = 8 \quad \therefore v_a = \frac{8}{7} = 1.14 \text{ V}$$

$$I_N = \frac{v_a}{1} = 1.14 \text{ A}$$

R_N :



KCL: $2i_x + \frac{v_a}{4} + \frac{v_a - 1}{1} = 0$

$$i_x = \frac{v_a}{4}$$

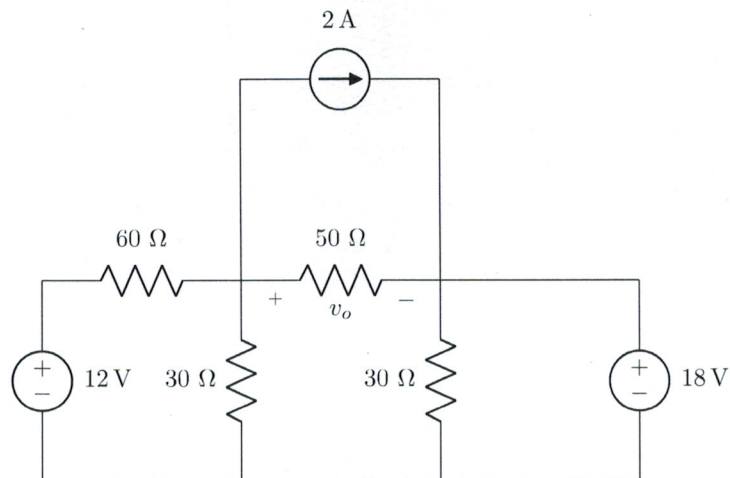
$$\therefore \frac{v_a}{2} + \frac{v_a}{4} + v_a - 1 = 0$$

$$2v_a + v_a + 4v_a - 4 = 0$$

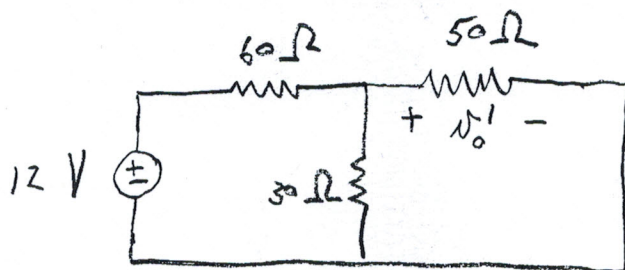
$$7v_a = 4 \quad \therefore v_a = 0.571 \text{ V}$$

$$i_o = \frac{1 - v_a}{1} = 0.429 \text{ A}$$

$$R_N = \frac{1}{i_o} = 2.33 \Omega$$

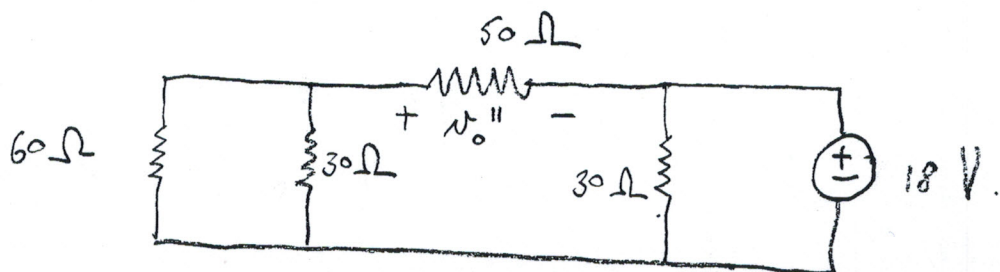
Questions 3 to 5 [3]Find the individual contributions of each of the 12 V, 30 V and 2 A sources to the voltage v_o .**03** Contribution of 12 V source [1]**04** Contribution of 18 V source [1]**05** Contribution of 2 A source [1]*See next page*

12 V :



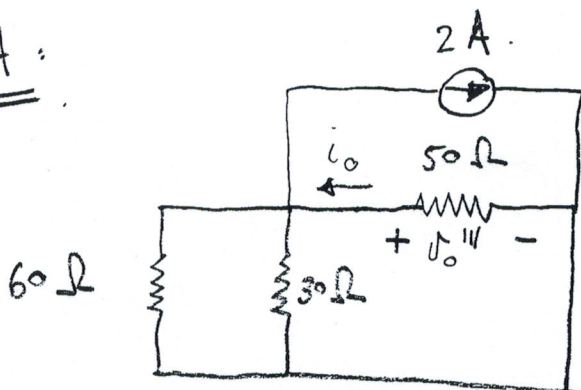
$$v_o' = 12 \frac{30 \parallel 50}{60 + 30 \parallel 50} = 2.86 \text{ V}$$

18 V :



$$v_o'' = 18 \frac{50}{60 \parallel 30 + 50} = 12.9 \text{ V}$$

2 A :

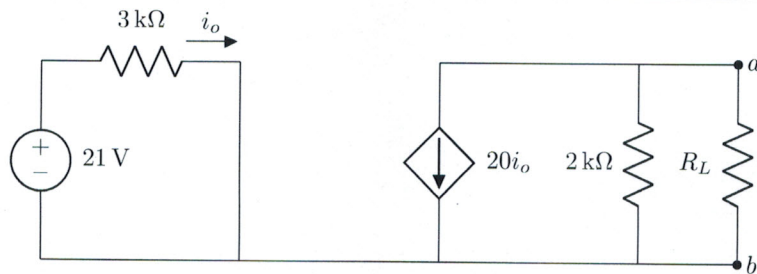


$$i_o = 2 \cdot \frac{30 \parallel 60}{30 \parallel 60 + 50} = 0.571 \text{ A}$$

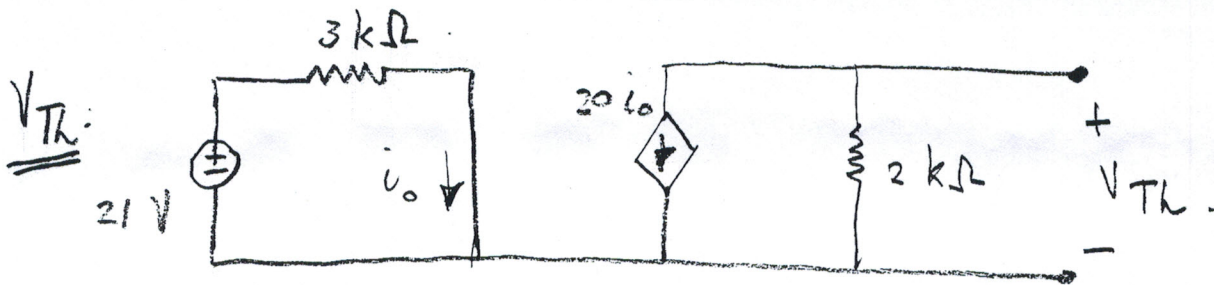
$$v_o''' = -(50)(0.571) = -28.6 \text{ V}$$

Questions 6 to 7 [2]

Find the Thevenin equivalent voltage V_{Th} seen by the load R_L , and the maximum power transferred to the load. It is known that the Thevenin equivalent resistance seen by the load is $2\text{ k}\Omega$.

06 V_{Th} [1]

07 Maximum power transferred to load [1]

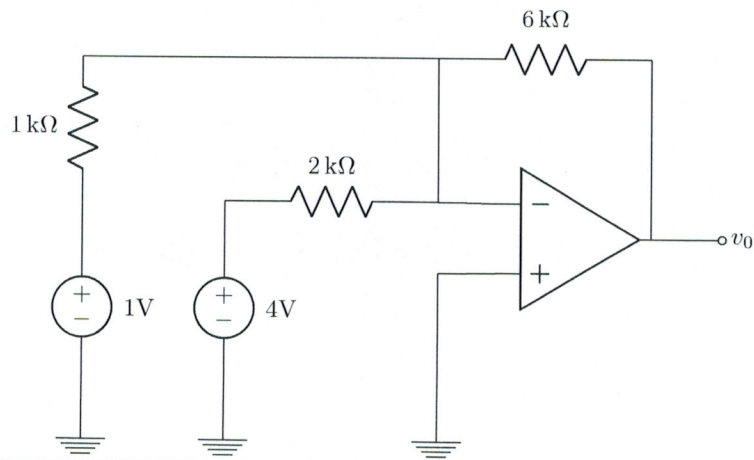


$$3000 i_o = 21 \quad \therefore i_o = 7 \text{ mA}$$

$$V_{Th} = -2000 (20 i_o) = -280 \text{ V}$$

Max power:

$$P_{max} = \frac{V_{Th}^2}{4 R_{Th}} = \frac{(-280)^2}{4 (2000)} = 9.8 \text{ W}$$

Question 8 [2]Find v_o . Assume that the op-amp is ideal.08 v_o . [2]

Summing amplifier:

$$\begin{aligned} v_o &= - \left(\frac{6000}{1000} (1) + \frac{6000}{2000} (4) \right) \\ &= - (6(1) + 3(4)) \\ &= -18 \text{ V} \end{aligned}$$

→

Questions 9 to 10 [3]

Two ideal op-amps are connected as shown. It is known that $v_1 = 60$ V. Find R_b and i_1 .

09	R_b [1]
10	i_1 [2]

left circuit: non-inverting op-amp

$$v_1 = 60 = \left(1 + \frac{R_b}{1000}\right) \cdot 20$$

$$3 = 1 + \frac{R_b}{1000}$$

$$2 = \frac{R_b}{1000} \quad \therefore R_b = 2 \text{ k}\Omega \quad \rightarrow$$

right circuit: inverting op-amp

op-amp
output:

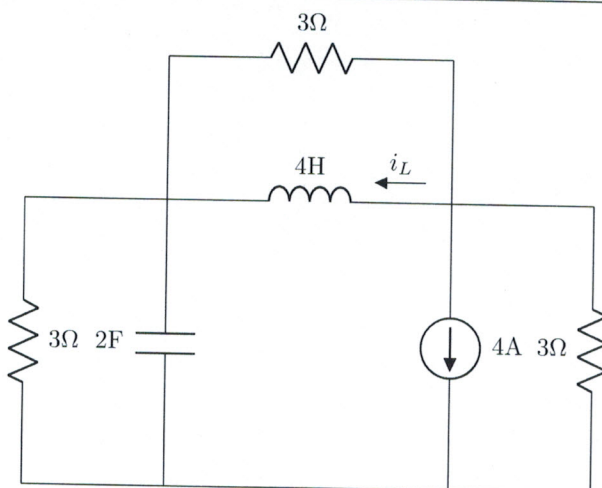
$$v_o = - \frac{18000}{6000} \cdot v_1$$

$$= -3 \cdot 60 = -180 \text{ V}$$

$$i_1 = \frac{v_o}{1000} = -0.18 \text{ A} \quad \rightarrow$$

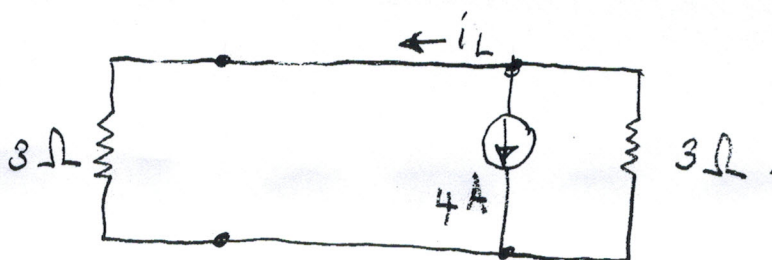
Questions 11 to 12 [2]

The circuit below operates under DC conditions. Calculate the current i_L , and the energy w_L associated with the inductor.



11 i_L [1]

12 w_L [1]



$$i_L = -4 \frac{3}{3+3} = -2 \text{ A}$$

$$W_L = \frac{1}{2} L i_L^2$$

$$= \frac{1}{2} (4) (-2)^2 = 8 \text{ J}$$